8	**A**	By Newton's 3rd law, the force acting on each particle is equal and opposite leading to change in momentum of each particle is equal and opposite. so, magnitude of $\Delta p_X =$ magnitude of Δp_Y $4 \times \Delta v_X = 9 \times \Delta v_Y$ $\Rightarrow \frac{\Delta v_X}{\Delta v_Y} = \frac{9}{4}$
9	**D**	$v = r\omega \quad \Rightarrow$ as r increases, v increases (since ω is constant)